



School of Technology

Program Guide for Students and Families

Michigan International Prep School | 2026–2027 Academic Year

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1. Welcome and Program Purpose

The School of Technology is where MIPS students study computer science, cybersecurity, IT, web development, and artificial intelligence, beginning as early as sixth grade.

Students write and run code, configure firewalls, explore virtual machines, train models, and build live websites.

A student who is curious about technology but unsure can find out here whether it suits them, while there is still time to try something else. A student who already knows can get a substantial head start: several semesters of coursework, a portfolio of finished projects, and in several courses, preparation for an industry certification exam.



What makes this program different

Professional tools and real projects. Several courses use the same training platforms and simulated labs that adults use to prepare for industry certifications. You configure firewalls, build virtual machines, write and run code, train models, and publish live websites. The games, programs, and sites you build are yours to keep.

Coordinated instruction across the program. School of Technology courses are coordinated by a program instructor who can help you connect your current work with previous courses and future pathways.

Courses that build on each other. Every course was chosen to fit a sequence. Section 4 lays out the five tracks and the pathway chart, so you can see where a course leads before you take it.

What you gain

The technical skills depend on which courses you take. Every pathway builds the same underlying habits: defining a problem clearly enough to solve it, breaking a large problem into smaller pieces, testing your own work and fixing what fails. You also practice explaining a technical idea to someone who does not already understand it, and thinking through how a technology affects the people who use it.

2. Who the Program Is For

Grade levels and prior experience

Courses are open to students in grades 6 through 12. The Foundations of Programming courses start at grade 6, most of the introductory courses open at grade 7, and the advanced, certification-aligned courses are limited to grades 9 through 12. The table in Section 4 gives the range for each course. You can start with no experience at all at any grade level.

This program is a good fit if you

- Are considering a technology field, or want to find out whether you should be.
- Want an industry certification, a portfolio you can show people, or a head start on college computer science.



- Are new to this, or already have experience. There is an entry point for beginners and enough depth in the advanced courses to challenge a student who arrives knowing some of it.

What the program asks of you

Students who do well here tend to share a few habits:

Ownership. The course supplies the sequence and the pacing targets. You supply the schedule. Deciding when in your week the work happens, and holding to it without a class period to prompt you, is the habit that separates students who finish from students who stall. It is also how technical work is organized after school. Developers, analysts, and IT professionals are given a goal and a deadline and are expected to manage the hours in between.

Determination. You will hit code that does not run and labs that do not pass on the first attempt. Technical work looks like that for professionals too. What gets you through is reading the error message, checking the instructions again, trying one more thing, and then asking for help with a specific question.

Curiosity. You want to know how the things you use every day actually work, and taking one apart sounds interesting rather than tedious. Ask why something works and you tend to come away understanding a layer of it that was invisible an hour earlier.

If any of these are areas you are still developing, start with a Foundations course and contact the instructor early so support can be established before pacing concerns grow.

3. How Learning Works

Course structure

One semester. Every course in the program runs a single semester. There are no year-long courses.

Asynchronous. There is no required class meeting time. Lessons, labs, readings, and assignments are available in the course platform, and you work through them on your own schedule.



Time commitment. Plan on about 45 minutes of course work every school day. The load varies by course, and the certification-aligned courses (PC Pro A and B, CyberDefense Pro, Artificial Intelligence) will ask more of you in some weeks than others.

What "at your own pace" means

You choose when during the week you work, and you may work ahead as far and as fast as you want. Each course has a pacing guide showing which unit you should have finished by which week.

There are no assignment deadlines and no late penalty. The weekly targets are recommended pacing benchmarks that keep the workload manageable. One date is fixed: everything must be turned in by the final day of the semester. Coursework is not accepted after that, and a course still incomplete on that date is graded as it stands.

Assignments, deadlines, and feedback

- Assignments are released and paced by unit. The pacing guide in your course shows the target completion date for each unit.
- Automatically graded work, including quizzes, lab simulations, and code checks, gives you results immediately. In most cases you can retry.
- Work that requires instructor review, such as projects and written responses, typically receives feedback within two school days.
- When a project comes back with revision comments, resubmitting it is part of the assignment.

Tech Talk Friday

Recommended, not required:

Grade band	Time (Fridays)
Grades 6–8	12:00–12:30 p.m.
Grades 9–12	1:00–1:30 p.m.

Current times are always posted in Google Classroom. You attend the session for your grade band regardless of which course you are taking, so students from across the program are in the room together. You are added to the meeting when you register for a course, and it appears on your calendar automatically.



These sessions are about community. Students talk about the projects they are working on, share what they have built, and dig into emerging technology with people interested in the same things. There is no direct instruction, and attendance is not required or graded.

How to get help

1. **Email or Google Chat the instructor.** This is the primary channel and usually the fastest. Expect a response within one school day.
2. **Request a one-on-one meeting during office hours.** Office hours are Monday, Tuesday, and Thursday from 9:00 to 10:30 a.m. You get a booking link when you enroll. This is the right channel for anything that needs screen sharing, a walkthrough, or a longer conversation.
3. **Bring it to the Friday video session.** Good for a quick question. For anything needing sustained attention, office hours are the better fit.

4. Technology Tracks and Course Pathways

Foundations of Programming

What you'll do: Write your first code, build games and animations, and compose and remix music with loops in Exploring Technology and Art. Move into text-based Python in Intro to Python by building projects like a Mad Libs generator, a quiz game, and a password authenticator.

Best fit: Students new to computer science, and anyone who wants a low-pressure way to find out whether they like programming before committing to a longer path.

Entry course: Either entry point—recommended sequence: Exploring Technology and Art → Intro to Python.

Where it leads: Web Design and Dev I, Artificial Intelligence, and college-level computer science. Programming skill also makes the Cybersecurity track significantly easier. Career connections include software development, data analysis, and game design, plus work that is not a technology job at all, from science to finance to music production. No certification exams are attached to these courses.



Cybersecurity

What you'll do: Build and defend a business network, configure firewalls and VPNs, and manage identity and encryption. Learn the professional process for testing systems, including reconnaissance, vulnerability scanning, and exploitation, along with the defensive side of honeypots and intrusion detection.

Best fit: Students drawn to the problem-solving and cat-and-mouse side of technology, and students curious about a field with a wide range of roles and entry points.

Entry course: Intro to Cybersecurity or Ethical Hacker. Either one opens the track, and either one satisfies the prerequisite for CyberDefense Pro.

Where it leads: CyberDefense Pro, and the IT and Hardware track, which covers the systems knowledge that security work sits on top of. Career connections include security analyst, SOC analyst, and incident response roles. CyberDefense Pro covers the content of the CompTIA CySA+ and TestOut CyberDefense Pro exams. See Section 6.

IT and Hardware

What you'll do: Learn how computers and networks actually work, install and troubleshoot hardware, work across Windows, Linux, and macOS, and build virtual machines. Move into managing a professional IT environment with Active Directory, scripting, and the command line.

Best fit: Students who like taking things apart and fixing them, and students who want coursework that connects directly to recognized industry certifications.

Entry course: IT Fundamentals. This is the one track that runs as a strict sequence: IT Fundamentals, then PC Pro A, then PC Pro B. Each prerequisite is required, not recommended.

Where it leads: The Cybersecurity track, which builds on the networking and systems knowledge from these courses. Career connections include help desk, IT support, and systems administration. IT Fundamentals covers the content of CompTIA Tech+, and PC Pro A and B cover CompTIA A+, a common entry credential for IT support work. See Section 6.



Web Development

What you'll do: Structure and style websites with HTML and CSS, publish a live portfolio homepage, and build responsive interfaces with Bootstrap and real UI and UX process.

Best fit: Students with a visual or design instinct, and students who want something they can show people immediately.

Entry course: Web Design and Dev I. Web Design and Dev II requires it.

Where it leads: Artificial Intelligence in the Emerging Technology track, and college coursework in computer science, design, or data. Career connections include front-end and web development and UI and UX design. No certification exams are attached to these courses.

Emerging Technology

What you'll do: Engineer prompts, train machine-learning models with TensorFlow and Google Colab, and work through the debates around deepfakes and ethics.

Best fit: Students who want to understand AI at a technical level instead of only using it.

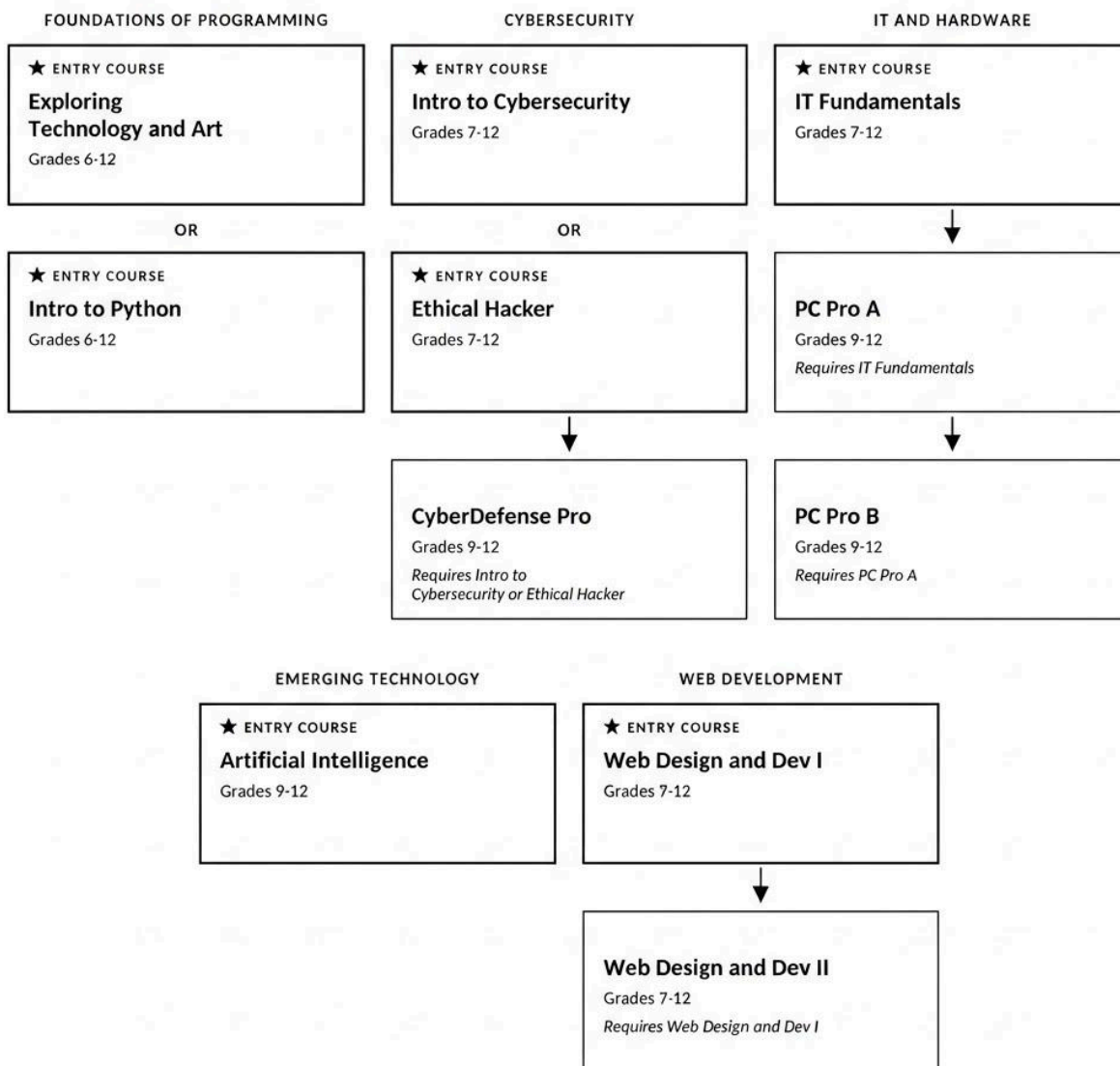
Entry course: Artificial Intelligence. This course is designed to be accessible for beginners and does not require any prior programming experience. If you are unsure where you stand, contact the instructor.

Where it leads: College coursework in computer science and data. Career connections include machine learning, data work, and any field putting AI tools into practice. No certification exam is attached to this course. This track holds one course today and will grow as new technologies are worth a full semester.



Pathway chart

School of Technology Course Pathways



Not every course runs every semester. See the course catalog at mipreptech.org for current offerings.



5. Student Expectations and Support

What successful participation looks like

Success depends on the ownership, determination and curiosity described in Section 2. In practice, that means:

Consistent engagement and pacing. You work in the course most weeks to stay at or ahead of the pacing guide. Courses built around labs and projects are difficult to complete in concentrated bursts and must be finished by the end of the semester.

Original work and honest use of AI. The code, labs, and writing you submit must be yours. Use AI to explain a concept, decode an error message, or review your own writing. It should not produce the work you hand in. The test is whether you can explain what you submitted and how it works. If you use an AI tool, disclose it and describe its role. Each syllabus states the specific AI policy for that course.

Seeking help and responding to feedback. Reach out while a problem is small. Feedback reaches you through your learning platform, Google Chat, or email. Successful students read these comments and revise their work, and revision is reflected in your grade.

Proactive communication. Email the instructor as soon as you anticipate a problem, such as illness or technical issues. Most challenges in this program are solvable when raised early.

Your weekly progress report

Every student in every course receives a progress report by email every Monday at 8:00 a.m. Mentors are copied.

What it shows	What it means
Overall grade	Your grade on the work you have completed so far. Missing assignments are not counted as zeros.
Relative grade	Your grade if the semester ended today and none of your remaining work was turned in. This is the number that tells you whether falling behind is starting to cost you.
Percent complete	How far through the course you are.



What it shows	What it means
Most recent activity	The last date you worked in the course, and what you were working on.
What comes next	Your next assignment, and the next major graded item ahead of you.

If you fall behind

If you fall more than three weeks behind the pacing guide, the instructor will contact you and your mentor to arrange a meeting. The purpose is to identify what is in the way and build a workable plan for the rest of the semester.

Some students deliberately front-load one course and complete another later in the semester. If that is your plan, arrange it with the instructor in advance. With an approved alternative pacing plan, your progress is evaluated against that plan.

What support you can count on

- The weekly progress report, sent to you with your mentor copied.
- Grades and completion percentage visible to you at all times in your course platform, alongside the pacing guide showing where you should be.
- Instructor support by email and Google Chat, including specific technical help on your actual code, labs, and configurations.
- The optional weekly group video session.
- One-on-one meetings by request, including screen sharing for technical troubleshooting.
- Written feedback on projects and instructor-reviewed assignments, with the opportunity to revise.
- Direct instructor contact for families and mentors who want a status update on a student.



6. Technology, Certifications, and Portfolio

What you need to participate

Requirement	Details
Device	A MIPS Chromebook and a reliable internet connection. A personal computer works as well.
Accounts	Your MIPS student account, plus the platform your course uses: CodeHS, TestOut/MICE, or Code.org. These are set up for you at enrollment, and there is nothing to buy.
Software	None to install. Every course runs in a browser. Use an up-to-date version of Google Chrome.
Google Classroom	Every course has one. It is used for announcements, links to course content, and student grades.

Certifications

Several courses cover the content of an industry certification exam:

Course	Related certification
IT Fundamentals	CompTIA Tech+
PC Pro A and PC Pro B	TestOut PC Pro, CompTIA A+
CyberDefense Pro	TestOut CyberDefense Pro, CompTIA CySA+

Completing a course does not itself earn the certification. The certification comes from passing the exam, and most students do additional study before they sit for it.



If you want to pursue one, tell the instructor. He can point you to the current exam objectives and practice materials, help you decide whether you are ready, and walk you through how and where to register.

MIPS does not provide certification exam vouchers. Registering and paying the exam fee are handled by you and your family. Fees vary by exam and change over time, so check the CompTIA or TestOut site for the current price before you plan around it.

Building a portfolio

What you make in these courses is yours to keep: a video game, a song you composed and remixed, a Python program, a live website, a working web application, or a documented network configuration. Save your work as you go. Web Design and Dev I includes building a live portfolio homepage, which gives you somewhere to collect everything else. A portfolio gives a college admissions reader or an employer more to look at than a line on a transcript.

7. How to Get Started

1. **Explore the tracks.** Look through the five tracks in Section 4 and the course listings on mipreptech.org. You are picking a direction to start in.
2. **Contact the instructor.** Email with your grade level, any prior technology experience, and what you are interested in. If you are unsure which course fits, this is the step that answers it. Students with prior experience should raise placement here.
3. **Talk with your counselor or mentor.** Your counselor or mentor confirms the course fits your schedule, your credit requirements, and your graduation plan.
4. **Verify prerequisites.** For PC Pro A, PC Pro B, CyberDefense Pro, and Web Design and Dev II, the instructor verifies you have completed the required course, or approves placement based on demonstrated experience. Artificial Intelligence has no required course, but check your programming readiness with the instructor before you request it.
5. **Submit the course request.** Your counselor enters the request through the normal MIPS registration process.
6. **Prepare before the semester begins.** Watch for your course platform account setup, your Google Classroom invitation, your office hours booking link, and the welcome message.



Placement based on demonstrated experience

If you have not taken the listed prerequisite but believe you are ready, you can ask the instructor to review your placement. Any of the following can support that request:

- **Prior coursework.** A comparable course taken at another school, a summer program, or an online course with a certificate of completion.
- **A certification.** A current industry certification covering the prerequisite material, such as CompTIA Tech+ for PC Pro A.
- **A project portfolio.** Work you built yourself, such as a published website, a GitHub repository, a home lab or virtual machine setup, or a game or application you can walk through and explain.
- **A readiness review with the instructor.** A short conversation, and in some cases a sample task, covering the skills the next course assumes. This is the fallback when the other three do not apply, and it is available to any student who asks.

Placement approval is a decision the instructor and your counselor make together.

8. Quick Answers

- **Do I need previous experience?** No. See *Section 2: Who the Program Is For*
- **How much time will this take?** Approximately 45 minutes per school day. See *Section 3: How Learning Works*
- **Does "at your own pace" mean no deadlines?** There are no individual late penalties, but all coursework must be completed by the final day of the semester.
- **How will I know if I am on track?** The instructor will give you weekly progress reports. See *Section 5: Student Expectations and Support*
- **Will my Chromebook work?** Yes. See *Section 6: Technology, Certifications, and Portfolio*
- **Do I have to attend the Friday sessions?** No. See *Section 3: How Learning Works*
- **Which courses have prerequisites?** See *Section 4: Technology Tracks and Course Pathways*
- **Will I earn a certification?** Maybe. See *Section 6: Technology, Certifications, and Portfolio*
- **How do I get help if I am stuck?** See *Section 3: How Learning Works*
- **Can I take more than one course, or change or drop one?** Usually yes, depending on your schedule and graduation plan. Talk to your counselor before adding, changing, or dropping



- **What if I do not finish by the end of the semester?** Missing work is scored as a zero in your final overall grade. *See Section 3: How Learning Works*
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9. Contact

Eric Muenchen

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Michigan International Prep School
muenchen@miprepschool.org

Website: mipreptech.org The course catalog, full course descriptions, syllabi, and current semester availability are all on mipreptech.org.

Office hours: Monday, Tuesday, and Thursday, 9:00–10:30 a.m., by appointment. Booking link: calendar.app.google/VwDT7bxiLayuJHww9

Your next step: Email Eric Muenchen with your grade level, any technology experience you have, and what you would like to learn. You will get a course recommendation back and can go from there.

For enrollment and scheduling questions, start with your MIPS counselor or mentor.